

```

5.  for name, local_gates in bell_states.items( ):
6.      qc = QuantumCircuit(2, 2)
7.      qc.h(0)
8.      qc.cx(0, 1)
9.      if "X" in local_gates:
10.         qc.x(1)
11.      if "Z" in local_gates:
12.         qc.z(1)
13.      qc.measure([0, 1], [0, 1])
14.      compiled = transpile(qc, sim)
15.      counts = sim.run(compiled, shots=1000).result(
16.          ).get_counts( )
17.      print(name, counts)

```

Now let us understand how the program works. Line 3 defines a dictionary whose keys name the four Bell states and whose values record the extra gates each one needs. Line 5 loops over them. Lines 6 to 8 build the $|\Phi^+\rangle$ circuit we already know, and Lines 9 to 12 apply the extra gates to qubit 1. Line 13 measures both qubits, and Lines 14 to 16 run the circuit for 1000 shots and print the counts. Remember to activate `qiskit_env` before running the program.

The results are shown in Figure 4, and the outcome is disappointing in a very instructive way. Both $|\Phi^+\rangle$ and $|\Phi^-\rangle$ produce roughly 500 counts of 00 and 500 of 11. Both $|\Psi^+\rangle$ and $|\Psi^-\rangle$ produce roughly 500 counts of 01 and 500 of 10. Four distinct, mutually orthogonal quantum states have collapsed into two indistinguishable pairs of statistics. The reason is arithmetic: a probability is the squared magnitude of an amplitude, and both $+1/\sqrt{2}$ and $-1/\sqrt{2}$ square to the same $1/2$. The minus sign is still in the state; it simply does not survive the squaring.

```

(qiskit_env) # python qiskit11.py
Phi+ {'11': 514, '00': 486}
Phi- {'00': 483, '11': 517}
Psi+ {'01': 520, '10': 480}
Psi- {'10': 495, '01': 505}

```

Figure 4: Output of the program `qiskit11.py`

Two orthogonal quantum states—as different as two states can be, and distinguishable in principle with perfect certainty—have been made to look identical simply by asking the wrong question. The information is not missing. We are asking in the wrong language.

So let us ask properly. Recall from the second article that quantum gates are reversible. If the circuit H followed by CNOT turns $|00\rangle$ into $|\Phi^+\rangle$, then running that circuit backwards—CNOT followed by H —must turn $|\Phi^+\rangle$ back into $|00\rangle$. And since the four Bell states form a basis, the same reversed circuit should map each of the other three to a

different computational basis state. This procedure is called a Bell measurement, and the program `'qiskit12.py'`, shown below, implements it.

```

1.  from qiskit import QuantumCircuit, transpile
2.  from qiskit_aer import AerSimulator

3.  bell_states = {"Phi+": "", "Phi-": "Z", "Psi+": "X",
4.                "Psi-": "XZ"}

5.  sim = AerSimulator( )

5.  for name, local_gates in bell_states.items( ):
6.      qc = QuantumCircuit(2, 2)
7.      qc.h(0)
8.      qc.cx(0, 1)
9.      if "X" in local_gates:
10.         qc.x(1)
11.      if "Z" in local_gates:
12.         qc.z(1)
13.      qc.cx(0, 1)
14.      qc.h(0)
15.      qc.measure([0, 1], [0, 1])
16.      compiled = transpile(qc, sim)
17.      counts = sim.run(compiled, shots=1000).result(
18.          ).get_counts( )
19.      print(name, counts)

```

Now let us understand how the program works. The only new statements are Lines 13 and 14, which apply a CNOT gate and then a Hadamard gate before the measurement in Line 15. That is the preparation circuit of Lines 7 and 8, run in reverse order. Everything else is unchanged.

The output is shown in Figure 5, and it is one of the most satisfying results we have obtained in this series. The state $|\Phi^+\rangle$ yields '00' in all 1000 shots. $|\Phi^-\rangle$ yields '01' in all 1000 shots. $|\Psi^+\rangle$ yields '10' every time, and $|\Psi^-\rangle$ yields '11' every time. Not approximately, not on average—every single shot. The randomness has vanished completely, and the four states that were indistinguishable a moment ago are now labelled with perfect certainty.

```

(qiskit_env) # python qiskit12.py
Phi+ {'00': 1000}
Phi- {'01': 1000}
Psi+ {'10': 1000}
Psi- {'11': 1000}

```

Figure 5: Output of the program `qiskit12.py`

Take a moment to appreciate what has happened, because it contains the seed of everything that follows in this series. The phase difference between $|\Phi^+\rangle$ and $|\Phi^-\rangle$ was invisible to measurement. By applying two more gates